LECTURER - DR. TRAN GIANG SON

Sleep Stage Classification from EEG Spectrograms

Group members

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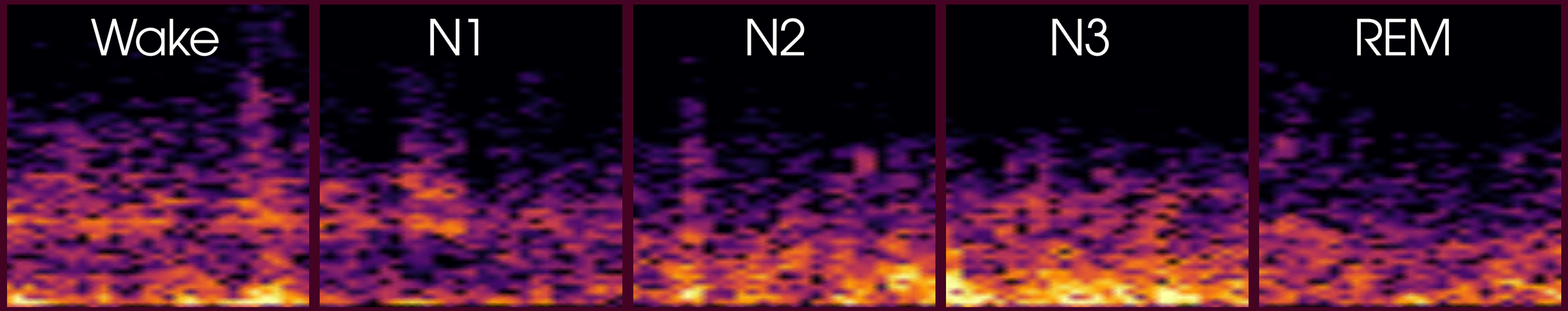
Problem Definition

- **Task**: Sleep stage classification
- **Input**: EEG spectrogram (30s)
- **Classes**: Wake / N1 / N2 / N3 / REM
- **Output**: sleep stage label



Example EEG Spectrogram Patterns for Sleep Stages

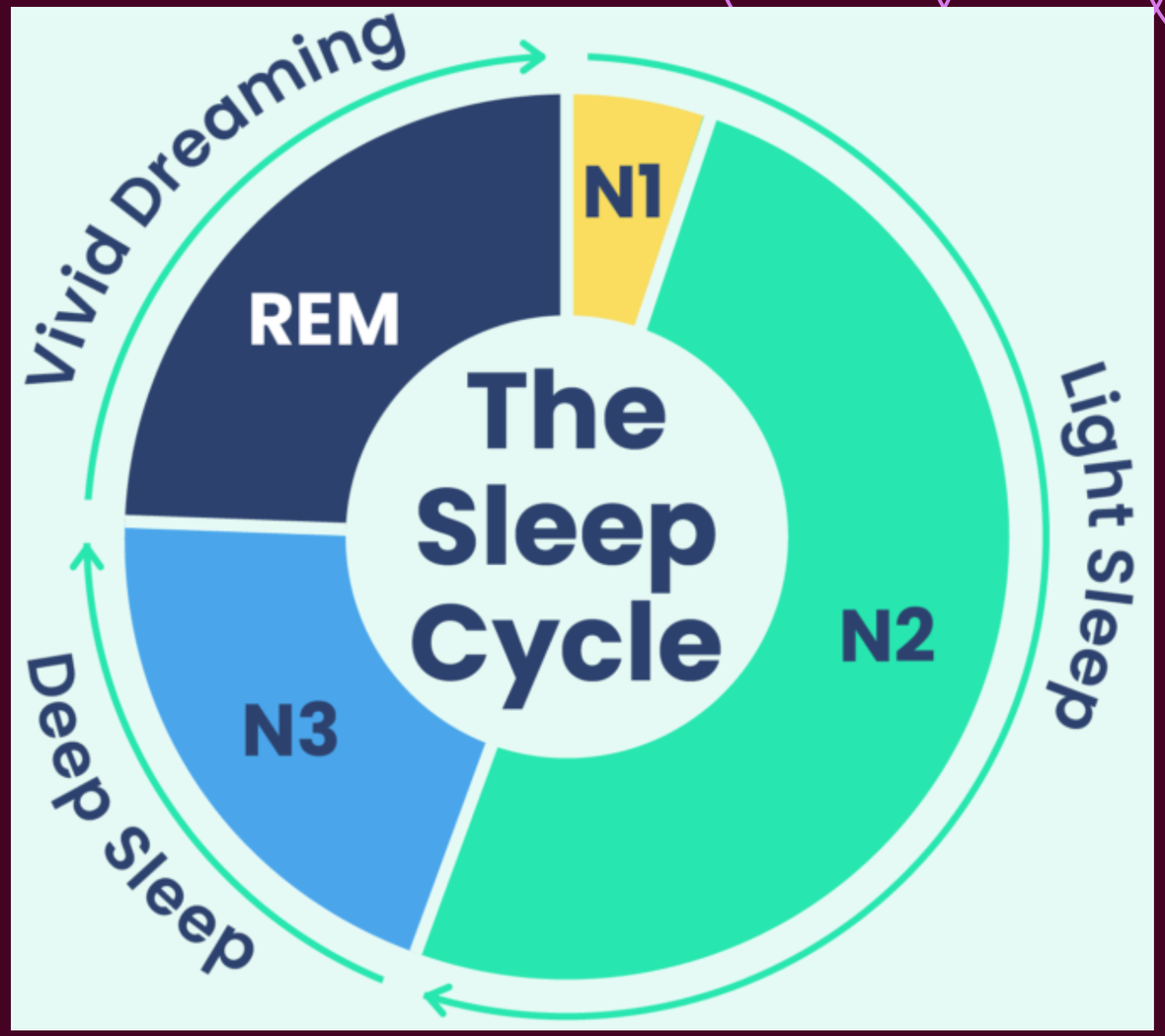
Motivation & Difficulties

Why Sleep Stage Classification?

- Sleep health importance
- Manual sleep labeling
- Requires sleep experts
- Large EEG data volume

Why is it difficult?

- Noisy EEG signals
- Similar sleep stage patterns
- Variability between patients
- Changes over time

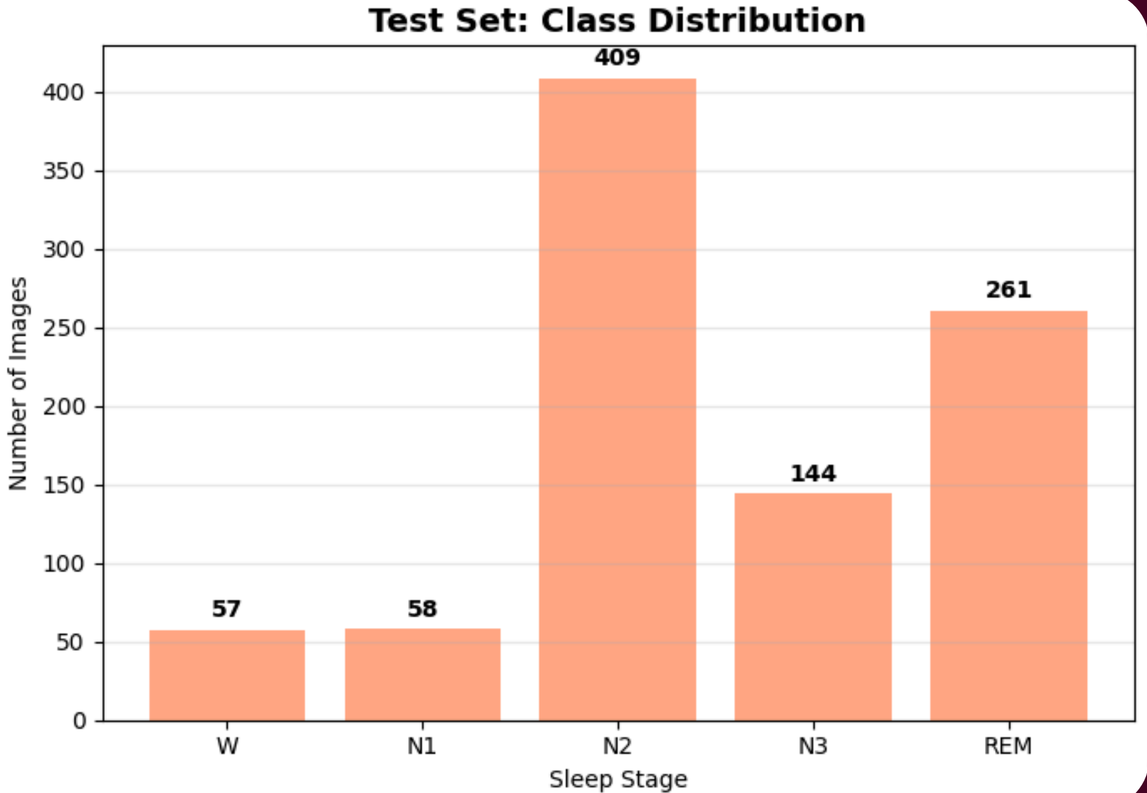


Sleep cycle stages during a typical night

DOD-H Sleep Spectrogram Dataset

- Input: EEG spectrogram images
- Segment length: 30 seconds
- Total samples: 5,578 images
- Classes: W / N1 / N2 / N3 / REM

- Delta wave (0.5-4Hz) : N3 (deepsleep)
- Theta wave (4-8Hz) : N1, REM
- Alpha wave (8-13Hz) : Wake, drowsiness
- Sleep spindles (11-16Hz) : N2
- Beta waves (13-30Hz) : Wake, REM



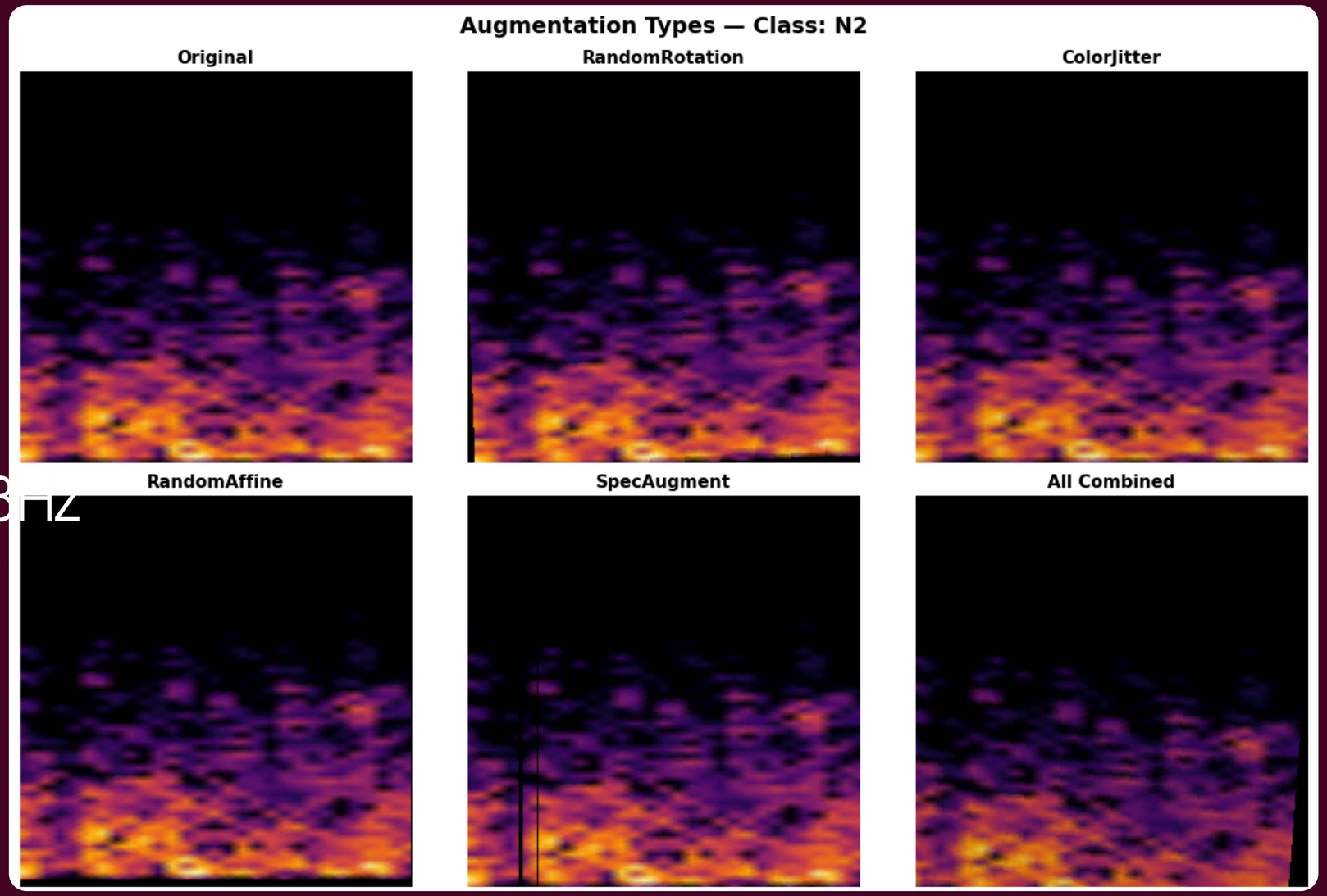
Data handling

Data augmentation

- RandomRotation
- RandomAffine
- ColorJitter

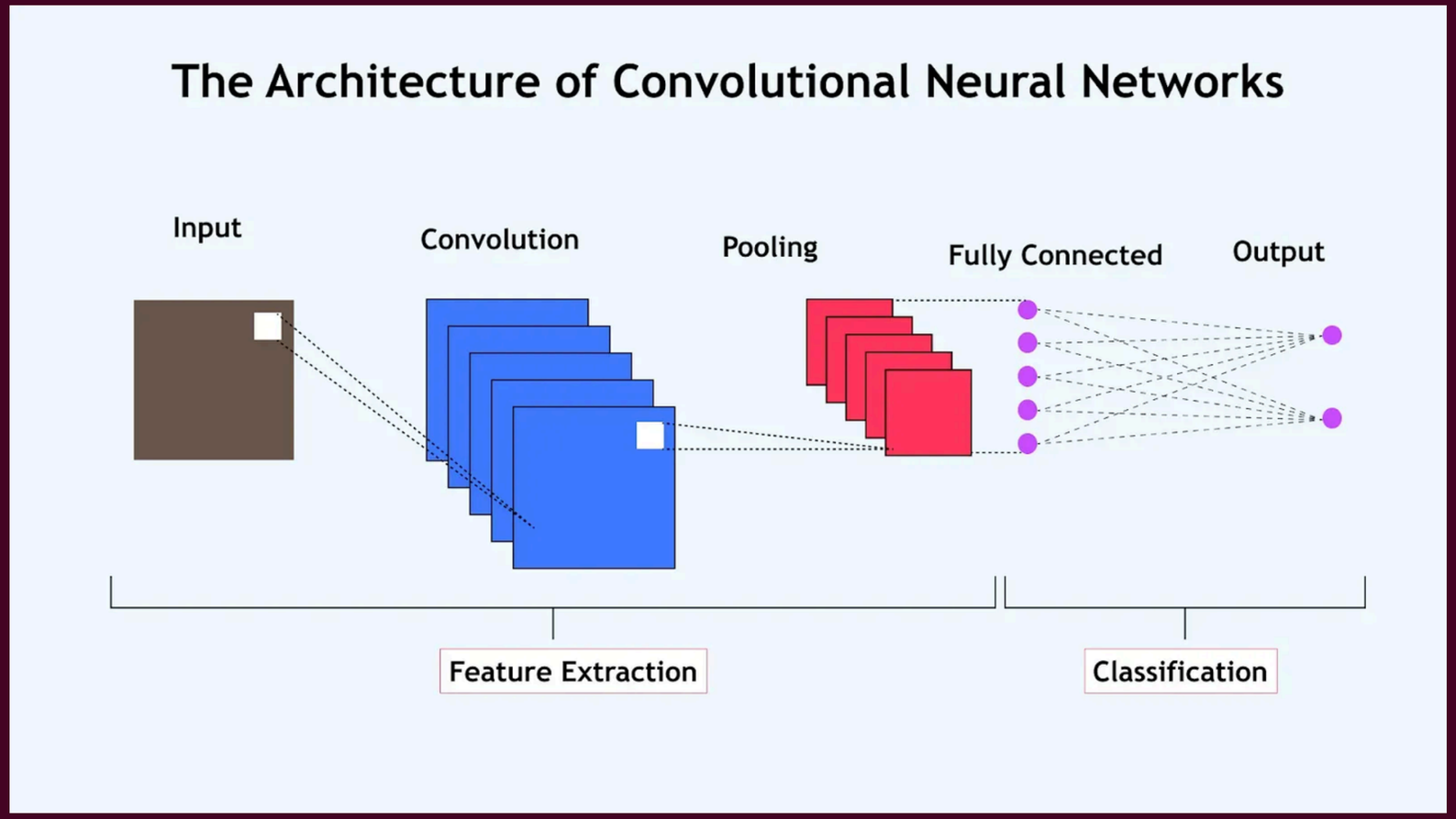
SpecAugment (1-2 masks)

- Frequency masking: 5-8 Hz
- Time masking : 3-5 time steps



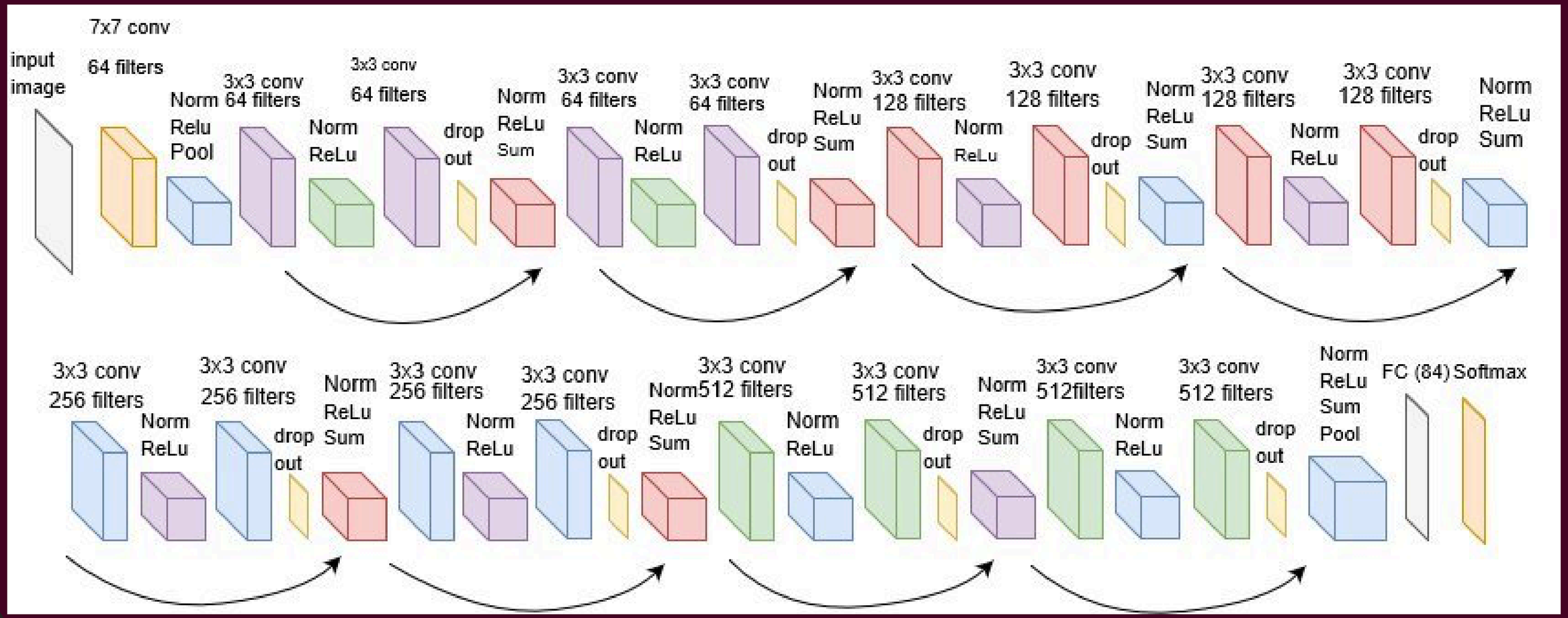
Custom - CNN

Stage	Layer	Channels
1	Conv + BN + ReLU + Maxpool	32
2	Conv + BN + ReLU + Maxpool	64
3	Conv + BN + ReLU + Maxpool	128
4	Conv + BN + ReLU + Maxpool	256
5	Conv + BN + ReLU + Adaptive AvgPool	512
Classifier	Linear + ReLU	256



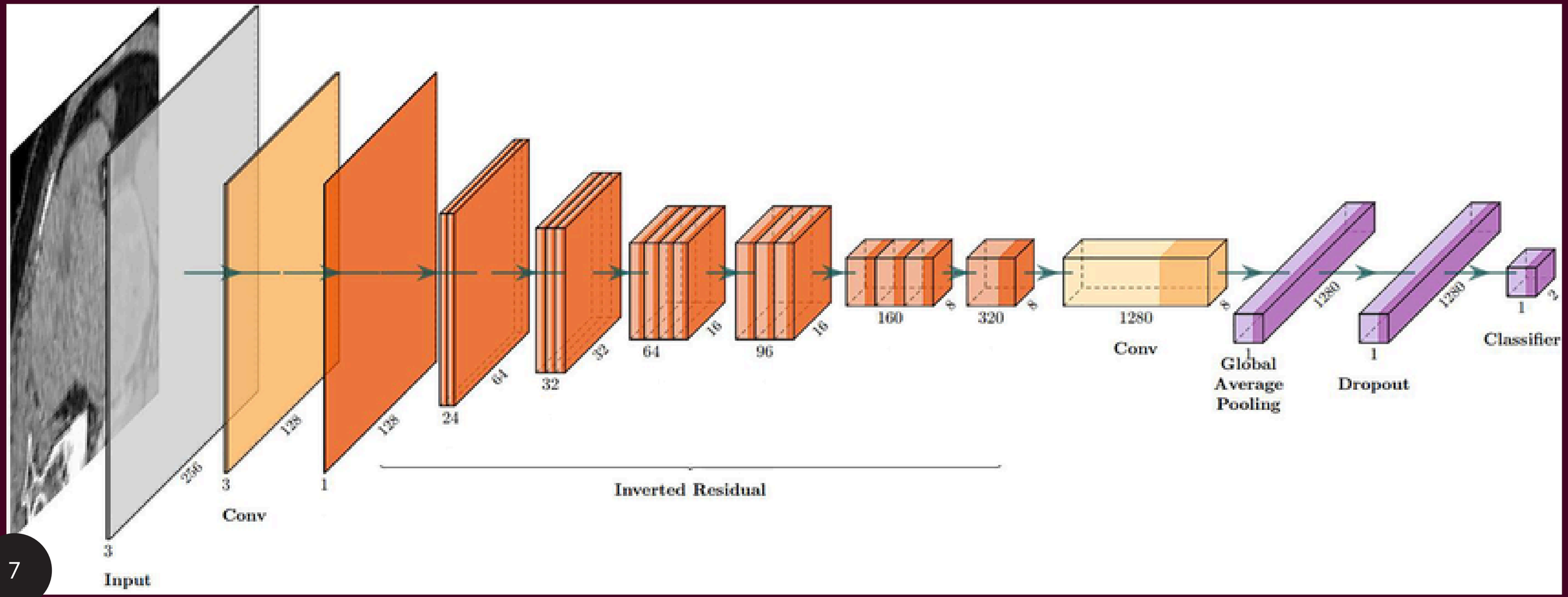
Resnet18

- Architecture Depth: 18 layers.
- Residual Learning: Skip connections
- Gradient Flow: Optimizes backpropagation
- Solving Degradation: Handles vanishing gradient problem / accuracy loss as depth increases.



MobileNetV2

- Bottleneck blocks learn visual features
- Global Average Pooling: feature maps -> vector

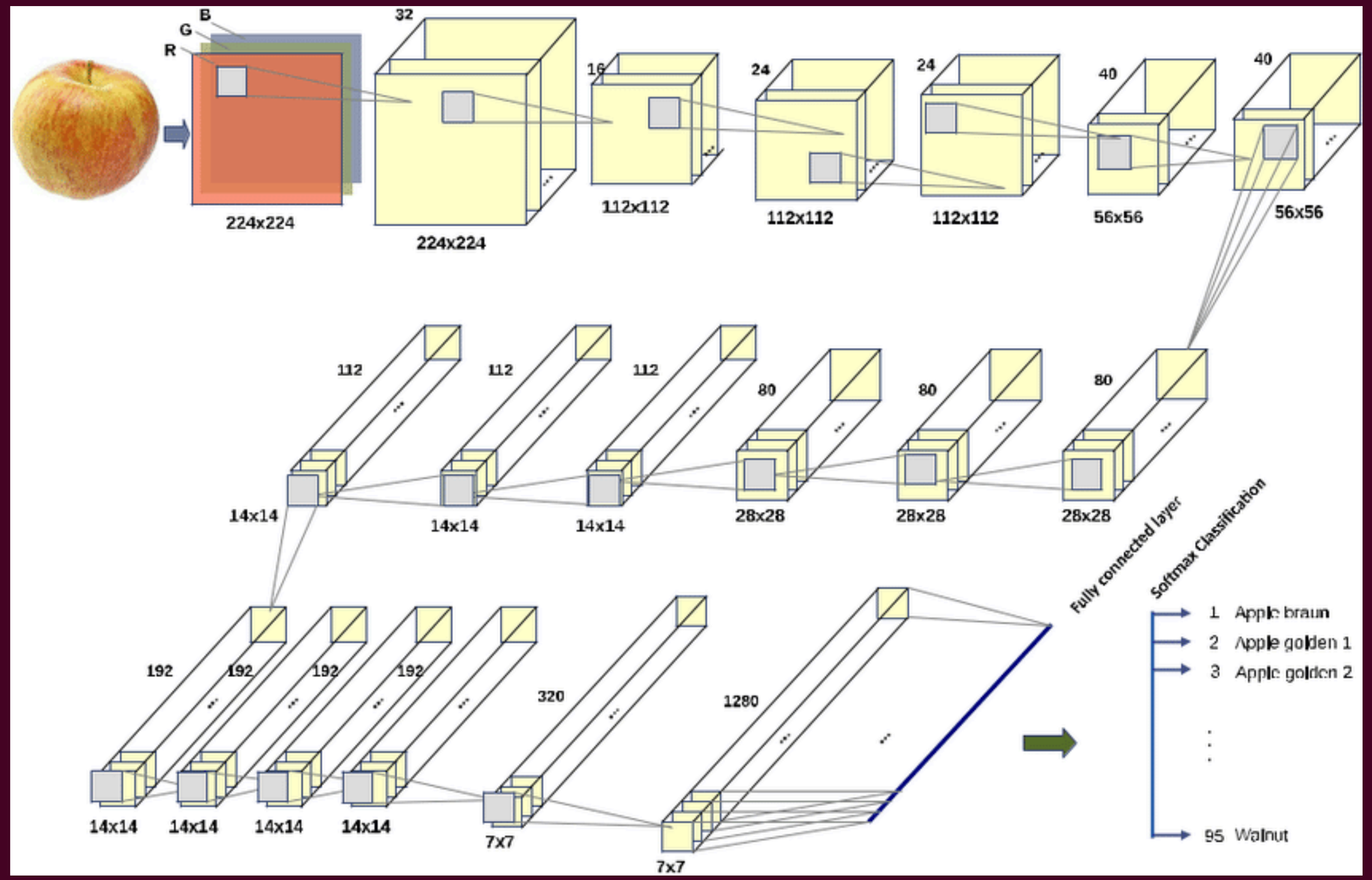


EfficientNet-B0

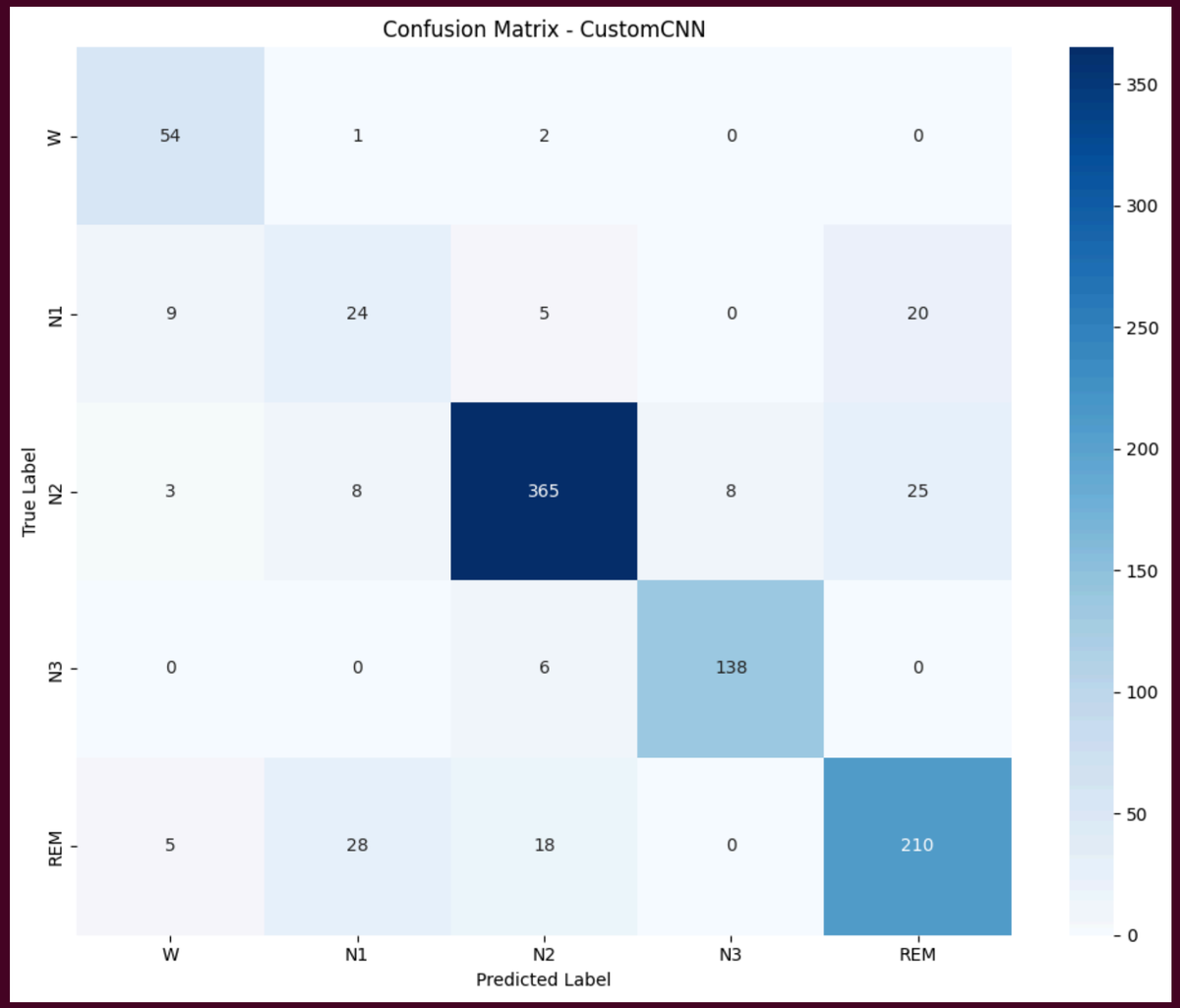
Compound Scaling:
- Balances depth, width, and input resolution

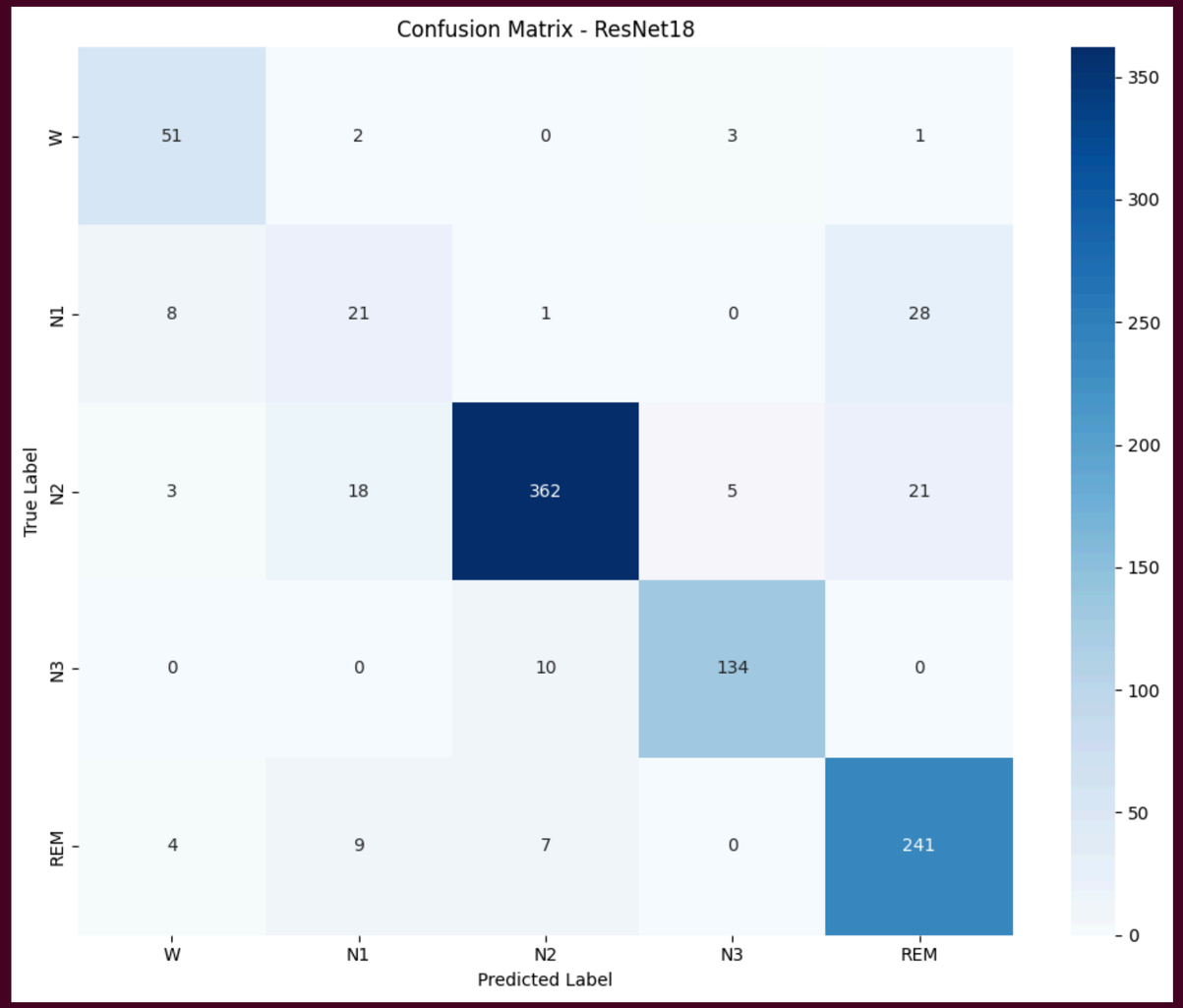
MBConv Blocks:
- MBConv with separable convolutions + Squeeze-and-Excitation attention

Feature Aggregation:
- Global average pooling converts feature maps



Confusion Matrixes





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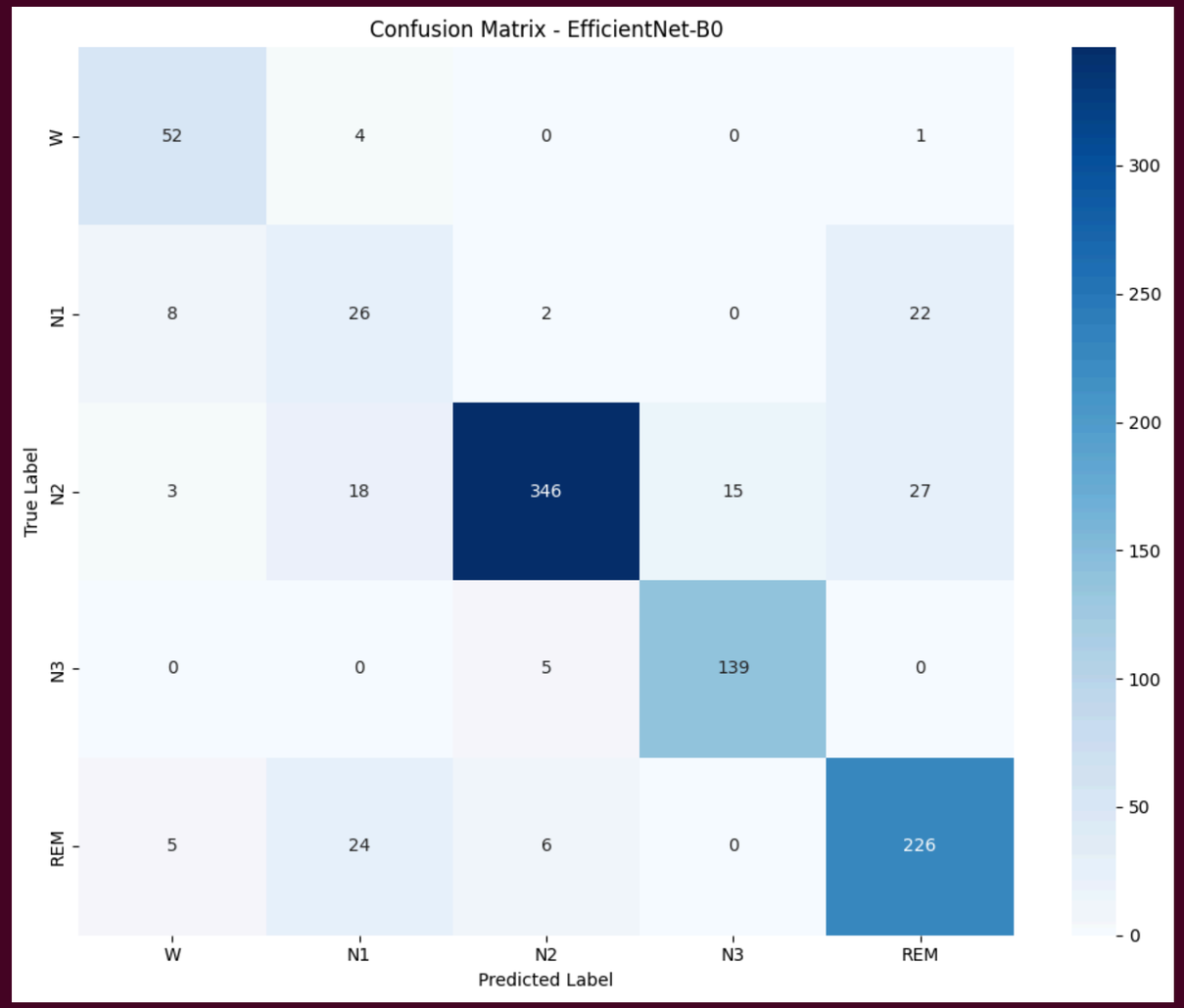
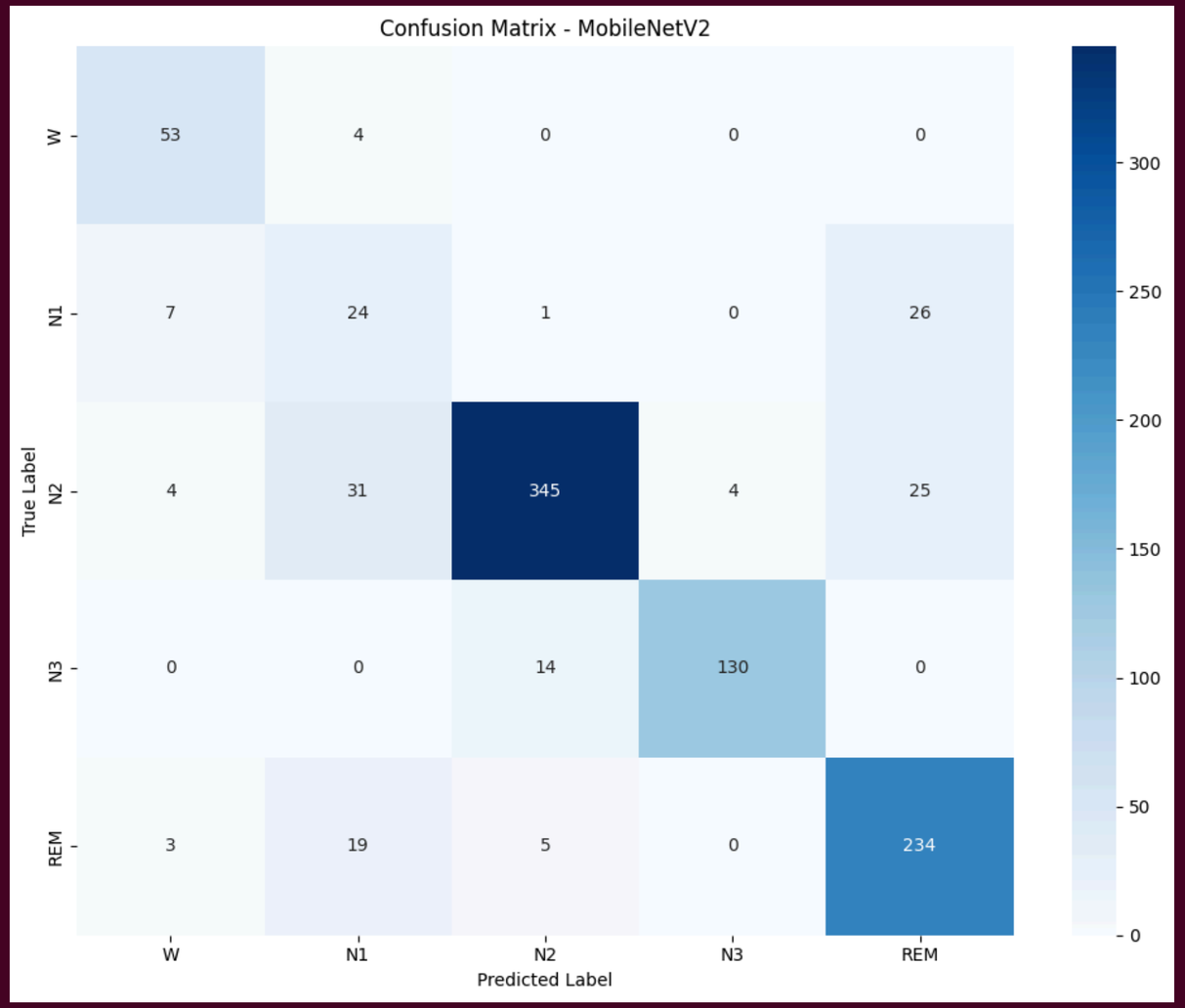
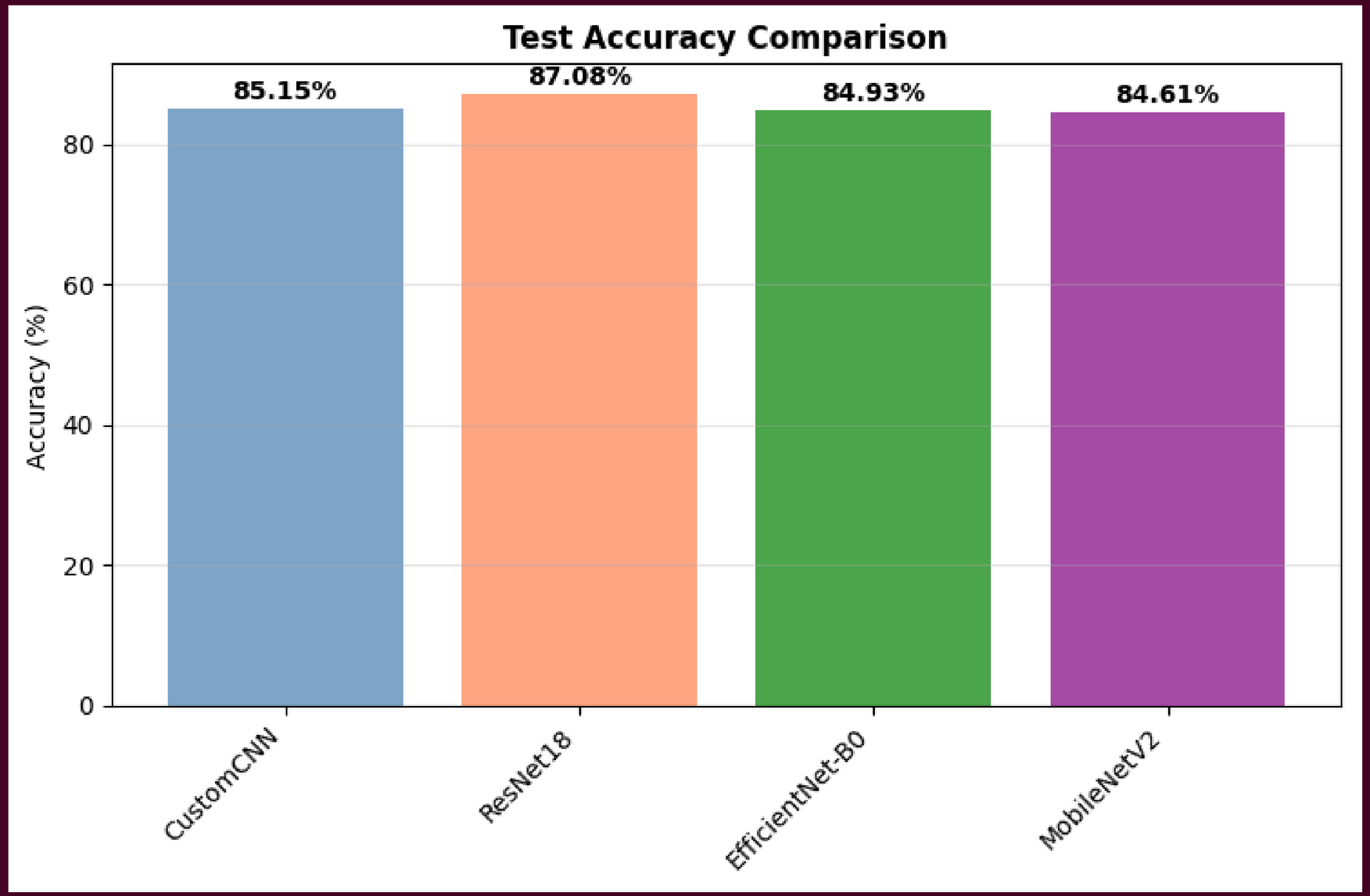




Evaluation methods

- Sensitive to class imbalance
- Rare stages ignored
- Not measure argeement with human
- Treats all errors equally

==> Cohen Kappa



Cohen Kappa Evaluation

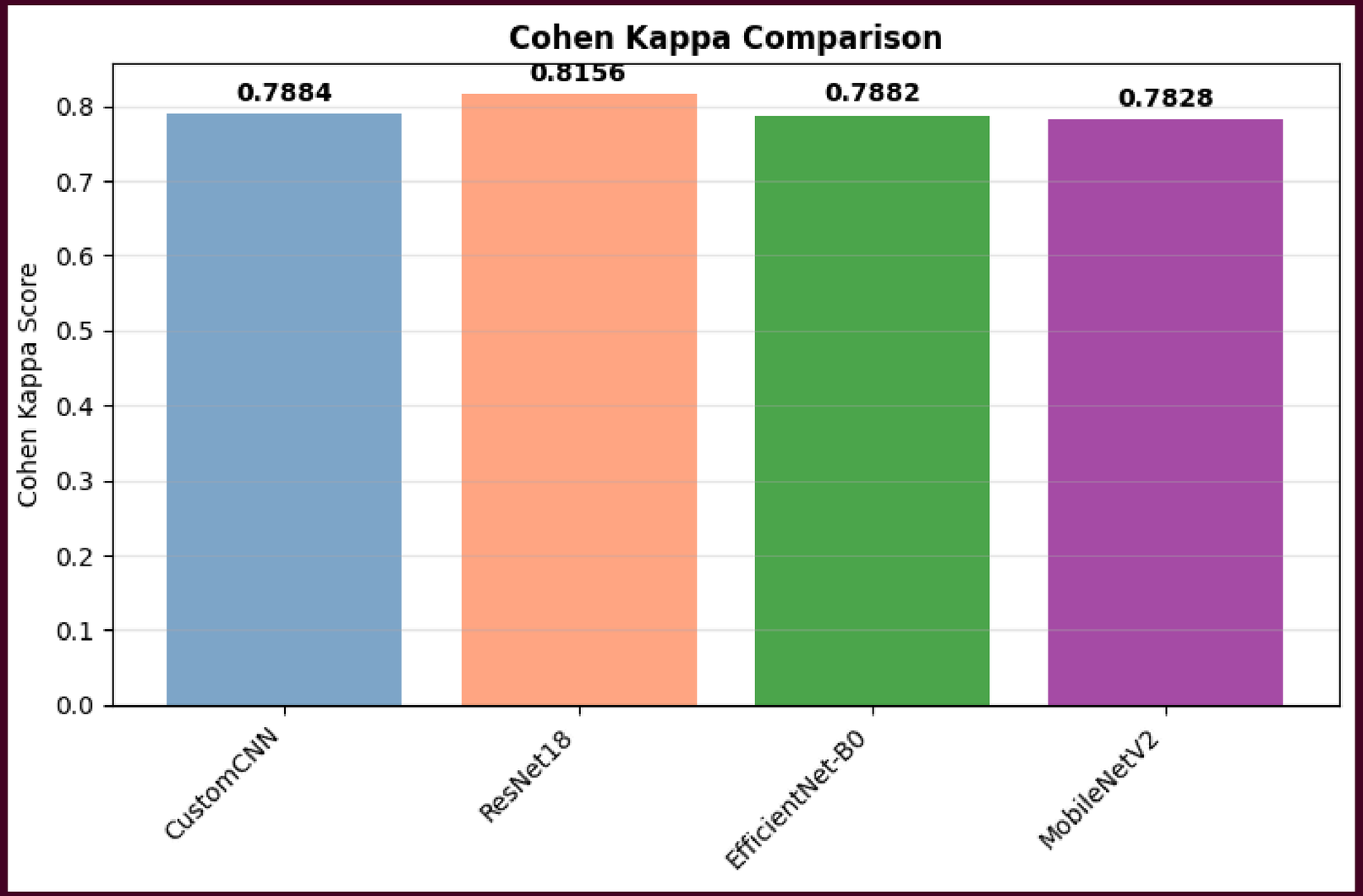
- Showing how better models are than random guessing
- Handle class imbalance (uneven distribution)

$$k = \frac{P_o - P_e}{1 - P_e}$$

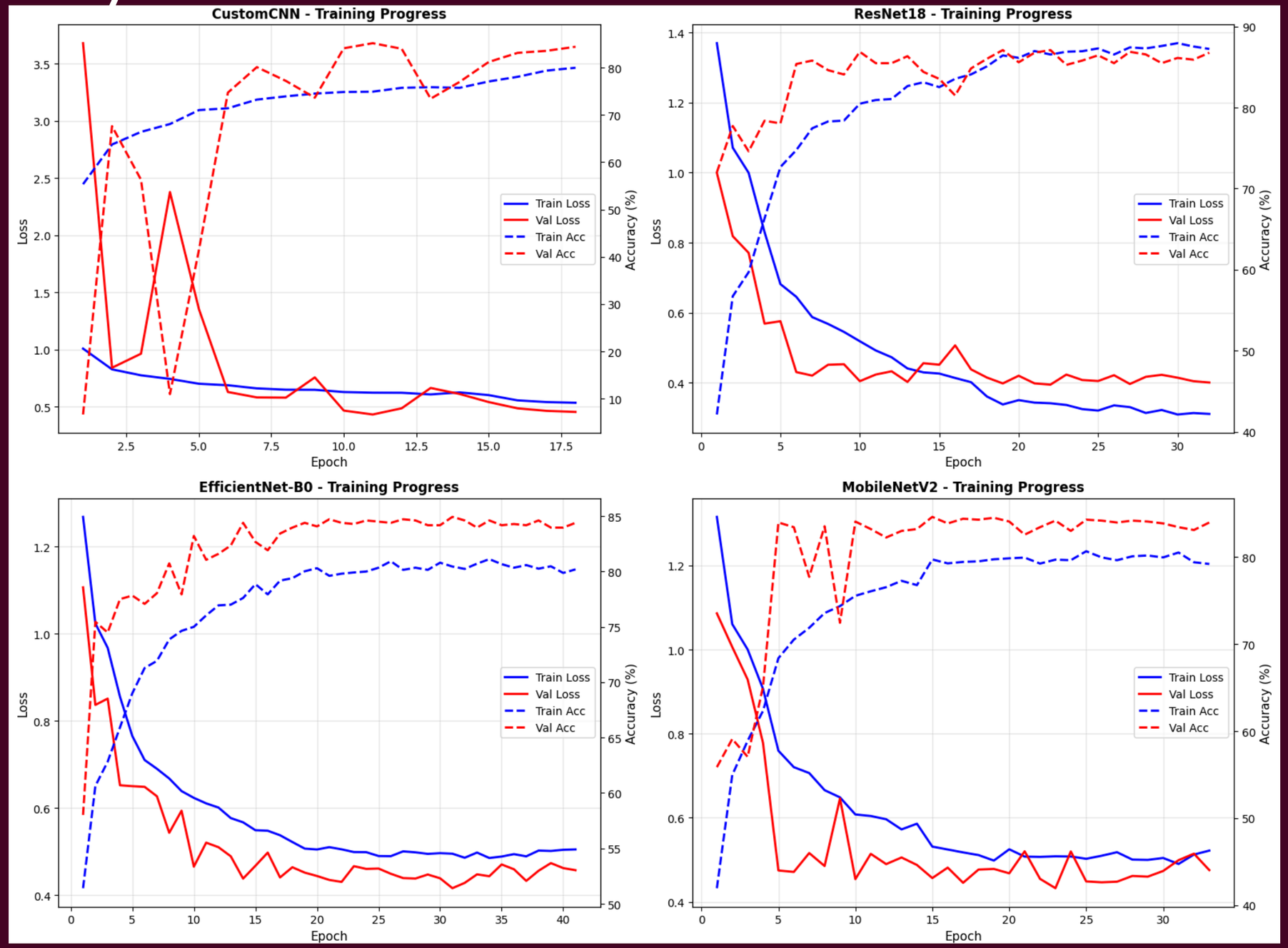
P_o : observed agreement

P_e : expected agreement by chance

$$P_e = \sum_{i=1}^{k} \left(p_i^{true} \cdot p_i^{pred} \right)$$



Accuracy and Loss cruve



Thanks for listening

Any questions?

Presented by Group 49